STATISTICS & MACHINE LEARNING · LESSON 7 OF 28

Introduction to Machine Learning: Types, Business Framing and the ML Lifecycle

What machine learning is and is not, how it differs from rules and statistics, supervised vs unsupervised vs reinforcement learning, regression vs classification vs clustering, turning a business problem into an ML problem, the end-to-end ML lifecycle (CRISP-DM to MLOps), linear vs non-linear models and link functions, and where GenAI fits.

ML classes 3–4 | ✦ scikit-learn 1.7+ · 2026 practice | Beginner | ⏱ 100 min

By the end of this lesson you can

- Define machine learning and distinguish it from rule-based systems and classical statistics
- Classify problems as supervised (regression, classification), unsupervised (clustering, dimensionality reduction) or reinforcement learning
- Translate a business question into target, features, unit of prediction, metric and decision
- Describe each stage of the ML lifecycle and common failure points
- Explain linear vs non-linear models, link functions, and when not to use ML

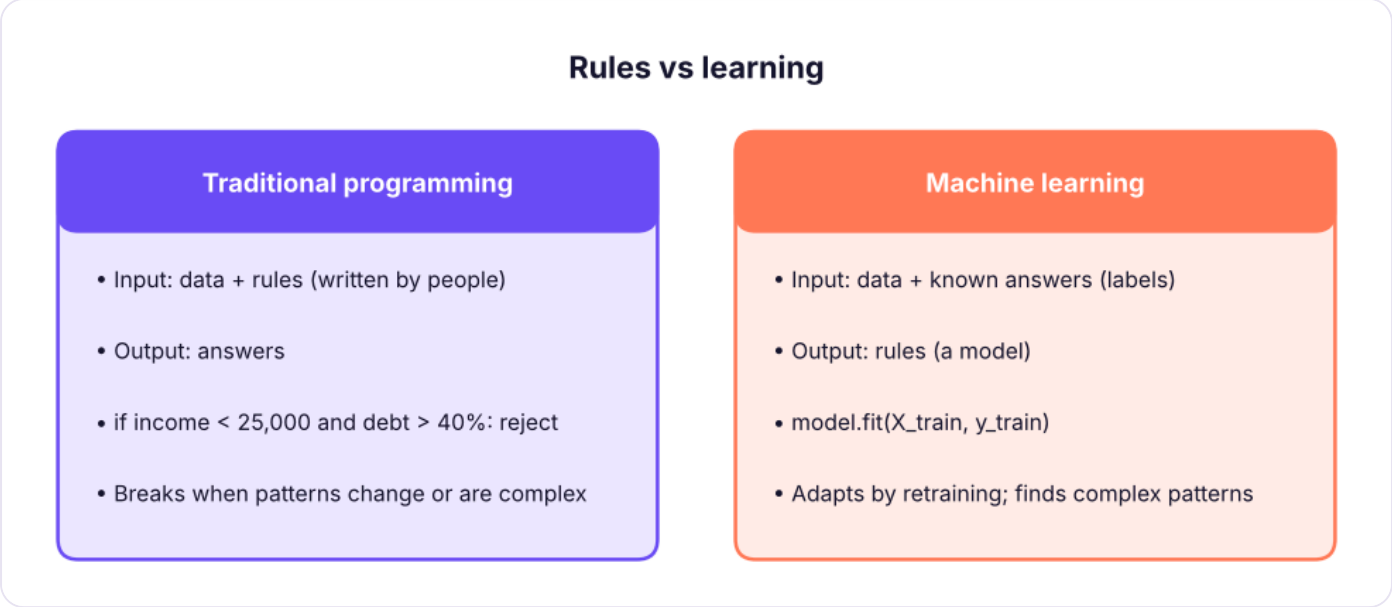
Datasets used

`house_prices.csv` `telecom_churn.csv`

What is machine learning?

> "A computer program is said to learn from experience E with respect to some task T and performance measure P if its performance at T, as measured by P, improves with experience E." (Tom Mitchell)

Instead of writing rules by hand, we give the computer **examples** (data) and let an algorithm **learn the rules** (a model) that map inputs to outputs.

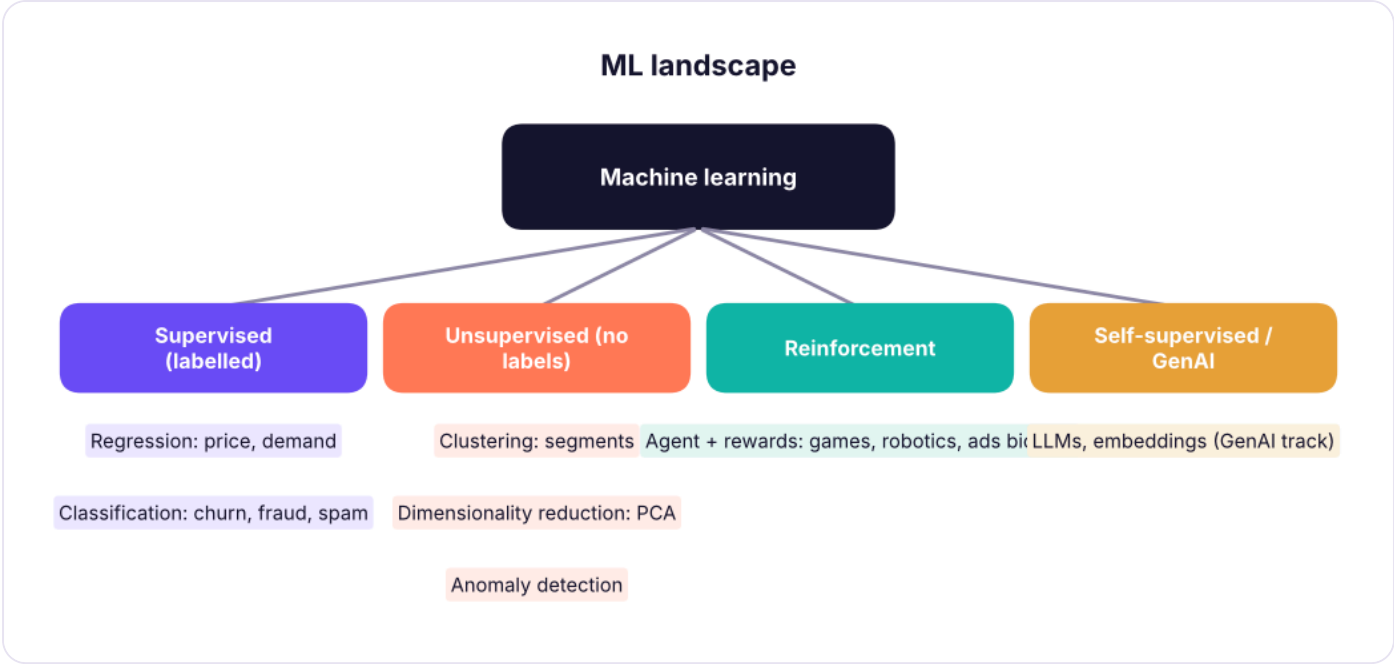


Traditional programming vs machine learning.

	Statistics	Machine learning
Main goal	Inference: understand relationships, test hypotheses	Prediction: accurate outputs on new data
Typical question	"Does overtime increase attrition, and by how much?"	"Which employees are likely to leave next quarter?"
Model choice	Interpretable, with assumptions	Whatever predicts best (with interpretability as needed)
Validation	p-values, confidence intervals	Held-out test data, cross-validation

They overlap heavily: linear and logistic regression belong to both.

Types of machine learning



The main families of machine learning.

Problem	Target type	Example	Algorithms (this track)
Regression	Number	House price, delivery time, demand	Linear, Ridge/Lasso, trees, boosting
Classification	Category	Churn yes/no, loan default, spam	Logistic, KNN, trees, RF, boosting, SVM, NB
Clustering	None	Customer segments	K-means, hierarchical, DBSCAN
Dimensionality reduction	None	Compress 50 features to 5	PCA
Time-series forecasting	Number over time	Next month's sales	Seasonal naive, ETS, ARIMA, ML with lags
Recommendation / ranking	Ranked list	Products to show	Collaborative filtering, learning to rank

Python

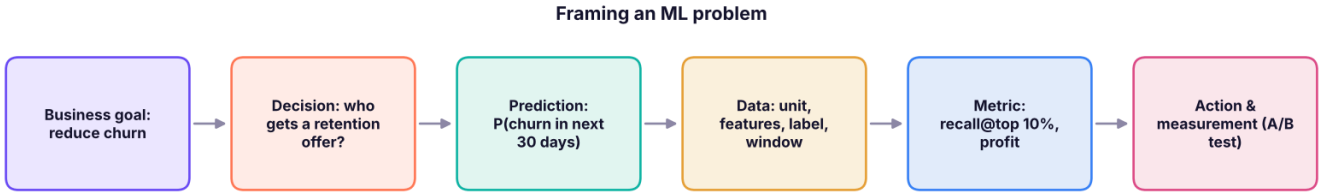
```
import pandas as pd
h = pd.read_csv("house_prices.csv"); tc = pd.read_csv("telecom_churn.csv")
print("Regression target:", "price_lakh", "| examples:", h.price_lakh.head(3).tolist(), "|
features:", [c for c in h.columns if c not in ("house_id", "price_lakh")])
print("Classification target: churn", tc.churn.value_counts().to_dict(), "| features:", [c
for c in tc.columns if c not in ("customer_id", "churn")])
```

OUTPUT

```
Regression target: price_lakh | examples: [60.2, 60.9, 43.7] | features: ['city',
'area_sqft', 'bedrooms', 'bathrooms', 'age_years', 'distance_to_center_km',
'parking']
Classification target: churn {'No': 2228, 'Yes': 772} | features: ['senior_citizen',
'tenure_months', 'contract', 'internet_service', 'payment_method', 'monthly_charges',
'total_charges', 'support_calls']
```

Vocabulary: a row is an **observation/sample**; input columns are **features** (X, predictors, independent variables); the column to predict is the **target** (y, label, dependent variable); the learned function is the **model**; using it on new data is **inference/prediction**.

From business problem to ML problem



Framing comes before algorithms.

Question	Churn example
What decision will the prediction change?	Which 10% of customers receive a ₹200 retention offer
Unit of prediction	One customer, scored monthly
Target definition	Cancelled within 30 days after the scoring date
Features (only information available **at prediction time**)	Tenure, contract, charges, support calls in the previous 90 days
Success metric	Recall in the top 10%, net profit of the campaign
Baseline	Current rule: target month-to-month customers with 3+ calls
Constraints	Explainability for the retention team, privacy, weekly batch scoring

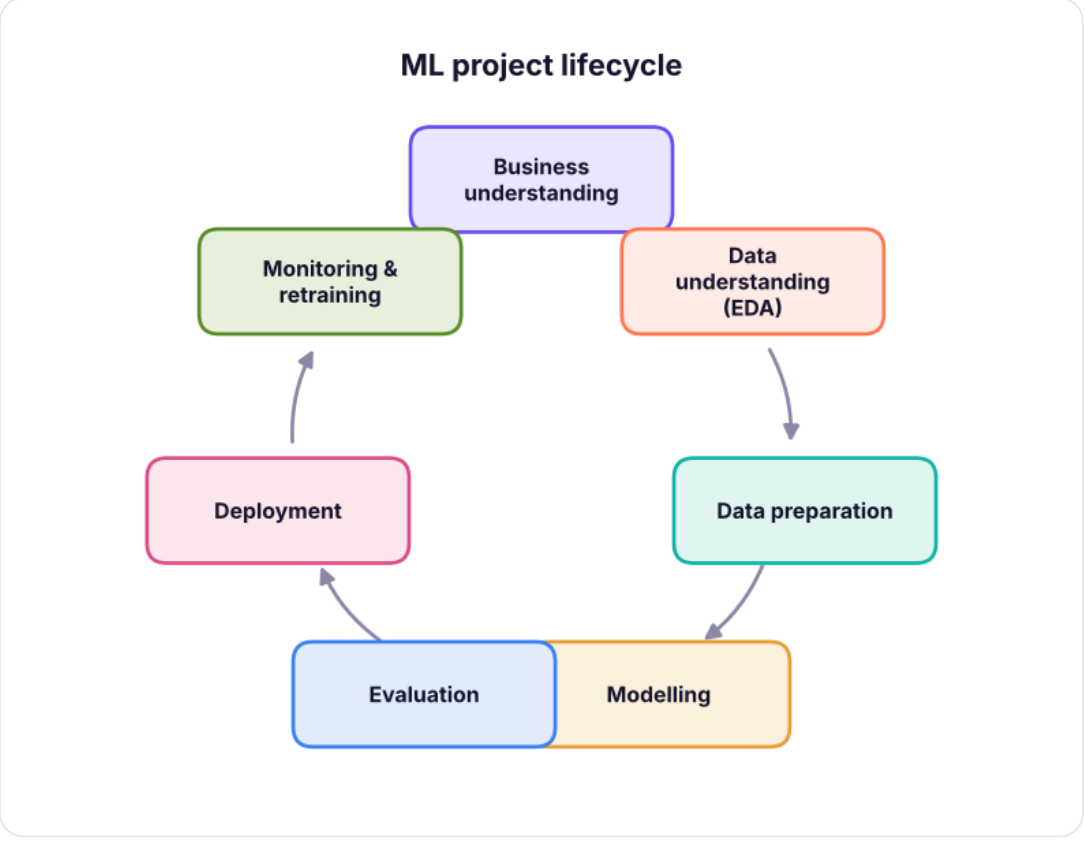
WATCH OUT **Data leakage**

Using information that would not be available at prediction time (for example "number of cancellation calls" recorded **after** the customer decided to leave) gives brilliant test scores and useless production models. Define a clear cut-off date for features and a later window for the label.

When not to use ML

Simple rules work; there is little data or no labels; mistakes are unacceptable and must be explained exactly; the pattern changes faster than you can retrain; or nobody will act on the predictions. Always compare with a simple baseline.

The ML lifecycle



CRISP-DM extended with MLOps: an iterative cycle, not a straight line.

Stage	Key work	Typical tools
Business understanding	Goal, decision, metric, baseline, constraints	Workshops, documents
Data understanding	Sources, EDA, quality, label definition	SQL, pandas, plots (Python track)
Data preparation	Cleaning, encoding, scaling, feature engineering, train/test split	pandas, scikit-learn pipelines
Modelling	Train candidate models, tune hyperparameters	scikit-learn, XGBoost, LightGBM
Evaluation	Test-set metrics, error analysis, fairness, business value	metrics, SHAP
Deployment	Batch scoring or API, integration with apps	FastAPI, Docker, cloud ML platforms
Monitoring	Data drift, performance decay, retraining	MLflow, Evidently, dashboards

In practice, most time goes to framing, data preparation and deployment, not to choosing algorithms.

Linear vs non-linear models and link functions

A **linear model** predicts with a weighted sum of features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ \dots \ + \beta_p x_p$$

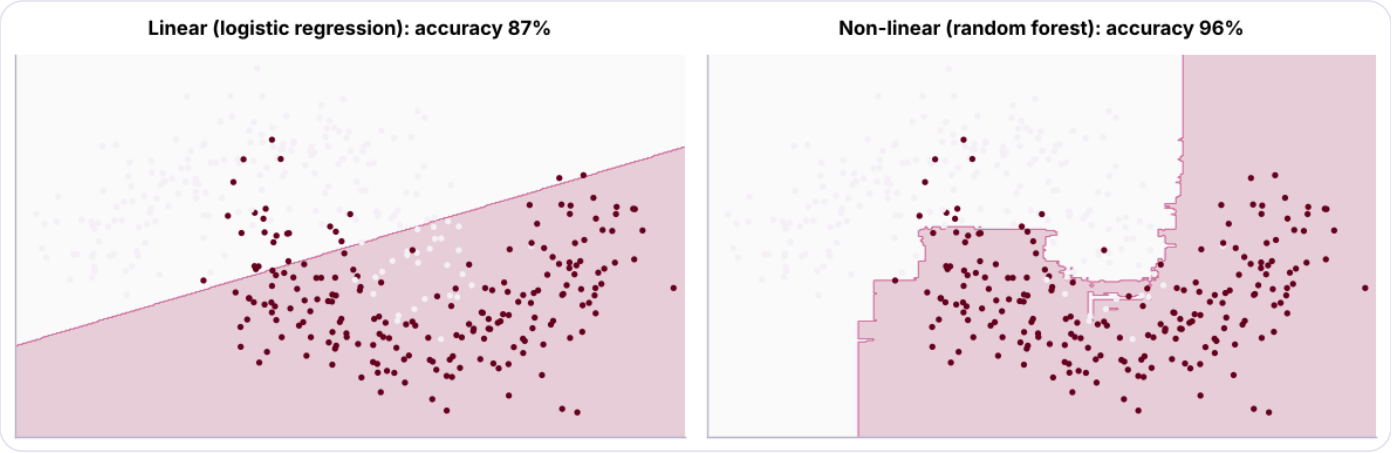
Generalised linear models pass that linear score through a **link function** so the output fits the target type:

Target	Model	Link (inverse)	Output
Any real number	Linear regression	Identity	ŷ = score
Probability (0–1)	Logistic regression	Logit ⇄ sigmoid 1/(1 + e^(−score))	P(y = 1)
Count (0, 1, 2…)	Poisson regression	Log ⇄ exp(score)	Expected count



The same linear score through three link functions: identity (regression), sigmoid (classification probability) and exp (counts).

Non-linear models (decision trees, random forests, gradient boosting, KNN, neural networks) capture thresholds and interactions automatically, usually at the cost of interpretability.



A linear model draws one straight boundary; a non-linear model can bend around the data.

Overfitting and generalisation (preview)

A model must work on **new** data. Memorising the training set (overfitting) fails in production; being too simple (underfitting) misses real patterns. We therefore always evaluate on data the model has not seen (lessons 8 and 15).

Where does GenAI fit?

Large language models are trained with self-supervised learning on huge text corpora, then adapted with instruction tuning and reinforcement learning from human feedback. In analytics they **complement** classical ML: LLMs read and write text, code and SQL; classical models remain best for structured, tabular prediction (churn, credit risk, demand), where gradient boosting usually wins. The GenAI track covers LLMs, RAG and agents.

Summary

KEY TAKEAWAYS

- ML learns rules from data; statistics focuses on inference, ML on prediction, with much overlap.
- Supervised (regression, classification), unsupervised (clustering, PCA, anomalies), reinforcement and self-supervised (LLMs) learning.
- Frame first: decision, unit, target and time windows, leak-free features, metric, baseline.
- Lifecycle: business → data → preparation → modelling → evaluation → deployment → monitoring, iteratively.
- Linear models use link functions (identity, sigmoid, exp); non-linear models capture thresholds and interactions.

Quick quiz

6 questions · tap an answer to check it

1. Predicting next month's sales in rupees is...

A Regression

B Classification

C Clustering

D Reinforcement learning

Answer: A. The target is a number.

2. Grouping customers into segments without labels is...

A Supervised learning

B Unsupervised learning (clustering)

C Regression

D A/B testing

Answer: B. No target column.

3. What is data leakage?

A Using information in training that would not be available at prediction time

B Losing data in a crash

C Sharing data publicly

D Missing values

Answer: A. It inflates test scores.

4. Which link function does logistic regression use to output a probability?

A Identity

B Sigmoid (inverse of the logit)

C Exponential

D Square root

Answer: B. It maps any score to 0–1.

5. Which stage often takes the most time in real ML projects?

A Picking the algorithm

B Data understanding and preparation

C Writing the final slide

D Installing Python

Answer: B. Data work dominates.

6. When might you NOT use ML?

A When a simple rule works well or nobody will act on predictions

B When you have lots of data

C When the pattern is complex

D When predictions save money

Answer: A. Start with a baseline.

EXERCISE

Frame these problems

For each, state the ML type, target, unit, two leak-free features, the metric and the action: (1) predicting loan default at application; (2) forecasting daily units for each store; (3) grouping 5,000 products by sales pattern; (4) flagging suspicious card transactions in real time.

Show solution (worked answer)

(1) Classification; default within 12 months; applicant; income, debt-to-income; recall/precision at the approval threshold and expected loss; approve/decline/price. (2) Regression/forecasting; units per store-day; store-day; lagged sales, calendar/holidays; MAPE/WAPE; stock orders. (3) Clustering; none; product; weekly sales profile, price; silhouette + business sense; assortment strategy. (4) Classification (imbalanced); fraud label; transaction; amount vs customer average, merchant risk; precision/recall at alert capacity; block or step-up authentication.

INTERVIEW CORNER

Questions you may be asked

Q: Explain machine learning to a non-technical person. Instead of writing rules ourselves, we show the computer many past examples with the right answers, and it learns patterns it can apply to new cases, like an employee learning from experience.

Q: Supervised vs unsupervised learning? Supervised learning uses labelled examples to predict a target; unsupervised learning finds structure such as groups or compressed representations without labels.

Q: How do you start an ML project? Clarify the business decision and success metric, define the target and prediction time, check data availability and leakage, set a simple baseline, then iterate on models.

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